Tính các tích phân bất định sau:

105. $\int \frac{xdx}{4+x^4}$

106. $\int \frac{dx}{x^2-x^4}$

107. $\int \frac{dx}{2^x+3}$

108. $\int \frac{\sin xdx}{\sqrt{4+\cos^2 x}}$

109. $\int \frac{x^2dx}{\sqrt{x^6+1}}$

110. $\int \frac{c^x dx}{\sqrt{c^{2x}-1}}$

111. $\int \sqrt{\frac{\ln\left(x+\sqrt{x^2+1}\right)}{x^2+1}}dx$

112. $\int \frac{\ln 2x}{\ln 4x}\frac{dx}{x}$

113. $\int \ln xdx$

114. $\int xarctgxdx$

115. $\int \frac{xarctgx}{\sqrt{1+x^2}}dx$

116. $\int c^{\sqrt{x}}dx$

117. $\int \frac{xc^x dx}{(x+1)^2}$

118. $\int \frac{\arcsin\sqrt{x}}{\sqrt{1-x}}dx$

119. $\int \frac{x^2dx}{\left(x^2+1\right)^2}$

120. $\int \frac{x-1}{x^2+x+1}dx$

121. $\int \frac{\ln xdx}{x\sqrt{1-4\ln x-\ln^2 x}}$

122. $\int \frac{5x^3+2}{x^3-5x^2+4x}dx$

Tính các tích phân xác định sau:

143. $\int_0^{\ln 2} \sqrt{e^x-1}\,dx$

144. $\int_0^1 \frac{\arcsin\sqrt{x}}{\sqrt{x(1-x)}}dx$

145. $\int_0^{\pi} \frac{x\sin x}{1+2\cos^2 x}dx$

146. $\int_0^3 \frac{dx}{\left(3+x^2\right)^{5/2}}$

147. $\int_0^{\pi/4} \cos^7 2xdx$

148. $\int_0^{\pi} \sin\frac{x}{2}dx$

149. $\int_0^{\pi/2} x\cos xdx$

150. $\int_1^{e^{\pi/2}} \cos\ln xdx$

151. $\int_{1/e}^{e} |\ln x|dx$

152. $\int_{-1}^{1} \frac{x^2\arcsin x}{\sqrt{1+x^2}}dx$

153. $\int_0^{\pi/3} \frac{x \sin x}{\cos^2 x} dx$

154. $\int_0^{\pi} (x \cos x)^2 dx$

155. $\int_{\pi/4}^{\pi/3} \frac{xdx}{\sin^2 x}$

156. $\int_1^{10} x \lg x dx$

157. $\int_0^1 \frac{\sqrt{x}-1}{\sqrt[3]{x}+1} dx$

158. $\int_0^{\pi/4} \frac{\sin^3 x dx}{\cos^8 x}$

159. $\int_0^{2\pi} \sqrt{1-\cos 2x}\ dx$

160. $\int_0^1 \frac{e^x dx}{e^{2x}+e^x+1}$

161. $\int_2^5 \frac{xdx}{\sqrt{5+4x-x^2}}$

162. $\int_0^2 |x-1| dx$

163. $\int_1^5 \frac{dx}{x+\sqrt{2x-1}}$

164. $\int_0^{\pi/2} \frac{dx}{3+2\cos x}$

165. $\int_0^{2\pi} \sqrt{1+\sin x} dx$

166. $\int_1^e x^2 \ln x dx$